

Air University



Operating Systems (Lab)

Year 2026

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: March 06, 2026

Lab Task # 05

Fork:

```
usama@ubuntu:~/Desktop$ vi program.c
usama@ubuntu:~/Desktop$ gcc program.c -o program
usama@ubuntu:~/Desktop$ ./program
--- Start ---
Hello from Process: 4725
Hello from Process: 4728
Hello from Process: 4729
Hello from Process: 4726
Hello from Process: 4727
Hello from Process: 4732
usama@ubuntu:~/Desktop$ Hello from Process: 4730
Hello from Process: 4731
```

Code:

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("--- Start ---\n");

    fork(); // 1 -> 2
    fork(); // 2 -> 4
    fork(); // 4 -> 8

    printf("Hello from Process: %d\n", getpid());

    return 0;
}
```

Execute:

```
usama@ubuntu:~/Desktop$ vi program2.c
usama@ubuntu:~/Desktop$ vi program1.c
usama@ubuntu:~/Desktop$ vi program2.c
usama@ubuntu:~/Desktop$ gcc program1.c -o program1
usama@ubuntu:~/Desktop$ gcc program2.c -o program2\
>
usama@ubuntu:~/Desktop$ gcc program2.c -o program2
usama@ubuntu:~/Desktop$ ./program1
PID of ex1.c = 4864
Back to ex1.cusama@ubuntu:~/Desktop$
```

Code:

```
#include <stdio.h>
#include <sys/types.h>
#include <stdlib.h>
#include <unistd.h>
int main(int argc, char *argv[])
{
    printf("PID of ex1.c = %d\n",
    getpid());
    char *args[] = {"Hello", "Neso",
    "Academy", NULL};
    execv("./ex2", args);
    printf("Back to ex1.c");
    return 0;
}
```

```
#include <stdio.h>
#include <sys/types.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
    printf("We are in ex2.c\n");
    printf("PID of ex2.c = %d\n",
    getpid());
    return 0;
}
```